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p. 476

leading coefficient of Φ being ε in place of 1, which introduces the factor ε^i , we have

$$\varepsilon^i \Psi_{m-i,0} = (-)^{\frac{(i-1)i}{2}} B'_i,$$

where B'_i is the Bezoutian secondary of the $(m-i-1)$ th degree in x to f and ϕ ; but B'_i has been proved $= B_i$, the Bezoutian secondary of the same degree to f and ϕ ; hence

$$\mathfrak{S}_{m-i,0} = (-)^{\frac{(i-1)i}{2}} B_i.$$

Art. (29). If now we return to the syzygetic equation,

$$\tau f - t\phi + \mathfrak{S} = 0,$$

$\mathfrak{S}$ may be treated as known, having in fact been completely determined as a function of the roots, as well in its most general form, as also so as to represent the simplified residues to f and ϕ in the preceding articles; it remains to determine the values of τ and t as functions of the roots corresponding to any allowable form of $\mathfrak{S}$, but I shall confine the investigation to the case where $\mathfrak{S}$ is the lowest-weighted conjunctive or, which is the same thing, a simplified residue to f and ϕ of any given degree in x ; each value of $\frac{\tau}{t}$ will then represent one of the convergents to $\frac{\phi}{f}$ when expanded under the form of a continued fraction. If $\mathfrak{S}$ be of the ι th degree in x , τ is of the degree $(n - \iota - 1)$ and t of the degree $(m - \iota - 1)$. This being supposed, and calling $n - \iota - 1 = \nu$, $m - \iota - 1 = \mu$, I say that t will be represented by G and τ by Γ , where

$$G = (-)^\iota \Sigma(x - h_{q_1})(x - h_{q_2}) \cdots (x - h_{q_\mu}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\mu} \\ \eta_1, \eta_2 \dots \eta_n \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\mu} \\ h_{\mu+1}, h_{\mu+2} \dots h_{q_m} \end{bmatrix}},$$

and τ is an analogous form Γ ; $h_1, h_2 \dots h_m$, as heretofore, being the roots of f , and $\eta_1, \eta_2 \dots \eta_n$ of ϕ . To fix the ideas and make the demonstration more immediately seizable, give m and n specific values; thus let $m = 5, n = 4, \iota = 2$, so that $\mu = 5 - 2 - 1 = 2$. Put $\mathfrak{S}$ under the form $\mathfrak{S}_{\iota,0}$, so that $\mathfrak{S}$ in the case before us

$$= \Sigma(x - h_{q_1})(x - h_{q_2}) \frac{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ h_{q_1} h_{q_2} \end{bmatrix}}.$$

Now make $x = h_1$, then $f = 0$, and $\mathfrak{S}$ becomes

$$\Sigma(h_1 - h_{q_1})(h_1 - h_{q_2}) \frac{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ h_{q_1} h_{q_2} \end{bmatrix}},$$

p. 477

that is

$$\Sigma \frac{\begin{bmatrix} h_1 \\ h_4 h_5 \end{bmatrix} \begin{bmatrix} h_1 h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_1 h_2 h_3 \\ h_4 h_5 \end{bmatrix}},$$

h_1 being kept constant in the above sum, but h_2, h_3, h_4, h_5 being partitionable in all the six possible ways into two groups, as into $h_4 h_5; h_2 h_3$ in the term above expressed. This sum is evidently identical with

$$\Sigma \frac{\begin{bmatrix} h_1 h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_1 h_2 h_3 \\ h_4 h_5 \end{bmatrix}}, \quad \text{that is} \quad \begin{bmatrix} h_1 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix} \times \Sigma \frac{\begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}}.$$

Again, ϕ becomes

$$\begin{bmatrix} h_1 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}.$$

Hence $t = \frac{\mathfrak{S}}{\phi}$ becomes

$$\Sigma \frac{\begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}}.$$

But, when $x = h_1, \frac{G}{(-)^t}$ becomes

$$\begin{bmatrix} h_1 \\ h_2 h_3 \end{bmatrix} \frac{\begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}},$$

that is

$$\frac{\begin{bmatrix} h_1 \\ h_2 h_3 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}} \begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix} = (-1)^t t.$$

Thus when $x = h_1$, $t = G$. In like manner, when $x = h_2$, or h_3 , or h_4 , or h_5 , t always $= G$; but t and G are both functions of x of the same, namely of only two, dimensions in x . Hence t is identical with G . So in general it may be proved that whenever $x = h_1$, or h_2 , or $h_3 \dots$ or h_m , t and G , which are each of only $(m - 1 - \iota)$ dimensions in x , are equal. Hence universally $t = G$, as was to be shown. To find τ we must avail ourselves of the symmorphic, or as we may better say (it being at the opposite extremity of the scale of forms), the antimorphic, value of $\mathfrak{S}$ represented by $\mathfrak{S}_{0,\iota}$, taking care to preserve $\mathfrak{S}$ strictly identical under both forms of representation, in point of sign as well as quantity. That is to say, we must make

p. 478

$$\begin{aligned}\mathfrak{S}_{0,\iota} &= (-)^{\iota(m-\iota)} \Sigma(x - \eta_{q_1})(x - \eta_{q_2}) \cdots (x - \eta_{q_\nu}) \frac{\left[\begin{array}{c} h_1, h_2 \dots h_m \\ \eta_{q_{\nu+1}}, \eta_{q_{\nu+2}} \dots \eta_{q_n} \end{array} \right]}{\left[\begin{array}{c} \eta_{q_{\nu+1}}, \eta_{q_{\nu+2}} \dots \eta_{q_n} \\ \eta_{q_1}, \eta_{q_2} \dots \eta_{q_\nu} \end{array} \right]} \\ &= (-)^\omega \Sigma(x - \eta_{q_1})(x - \eta_{q_2}) \cdots (x - \eta_{q_\nu}) \frac{\left[\begin{array}{c} \eta_{q_{\nu+1}}, \eta_{q_{\nu+2}} \dots \eta_{q_n} \\ h_1, h_2 \dots h_m \end{array} \right]}{\left[\begin{array}{c} \eta_{q_{\nu+1}}, \eta_{q_{\nu+2}} \dots \eta_{q_n} \\ \eta_{q_1}, \eta_{q_2} \dots \eta_{q_\nu} \end{array} \right]},\end{aligned}$$

where

$$\omega = \iota(m - \iota) + m(n - \iota),$$

so that

$$(-)^\omega = (-)^{\iota-1+mn-m\iota} = (-)^{mn-\iota};$$

and consequently the same reasoning as was applied to t to prove $t = G$, will serve to show that $-\tau = \Gamma$, where

$$\Gamma = (-)^{mn} \Sigma(x - \eta_{\xi_1})(x - \eta_{\xi_2}) \cdots (x - \eta_{\xi_\nu}) \frac{\left[\begin{array}{c} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_\nu} \\ h_1, h_2 \dots h_m \end{array} \right]}{\left[\begin{array}{c} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_\nu} \\ \eta_{\xi_{\nu+1}}, \eta_{\xi_{\nu+2}} \dots \eta_{\xi_n} \end{array} \right]},$$

or

$$\tau = (-)^\omega \Sigma(x - \eta_{\xi_1})(x - \eta_{\xi_2}) \cdots (x - \eta_{\xi_\nu}) \frac{\left[\begin{array}{c} h_1, h_2 \dots h_m \\ \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_\nu} \end{array} \right]}{\left[\begin{array}{c} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_\nu} \\ \eta_{\xi_{\nu+1}}, \eta_{\xi_{\nu+2}} \dots \eta_{\xi_n} \end{array} \right]},$$

where

$$\omega = mn - 1 - m\nu = mn - 1 - m(n - \iota - 1) = m\iota + m - 1.$$

Art. (30). I have not succeeded in throwing t and τ under any other than the single forms for each above given, and it is remarkable that whilst apparently t and τ admit only of this single representation, $\mathfrak{S}$ admits of the variety of forms included under the general symbol $\mathfrak{S}_{\nu-\iota}$ for a given value of ι ; and it ought to be remarked that these forms, although the most perfectly symmetrical and exactly balanced representations, and for that reason possibly the most commodious for the ascertainment of the allotrious factor belonging to them respectively, by no means exhaust the almost infinite variety of modes by which the simplified residues, that is, the hekistobarytic, or if we like so to call them, the prime conjunctives, admit of being represented as functions of the roots of the given functions; for if in Art. 16, instead of writing

$$R = \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \end{bmatrix} \text{imes} \begin{bmatrix} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{bmatrix}},$$

p. 479

we had made

$$R = \frac{P(h_{q_1}, h_{q_2} \dots h_{q_v}; \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}})}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_v} \\ h_{q_{v+1}}, h_{q_{v+2}} \dots h_{q_m} \end{bmatrix} \text{imes} \begin{bmatrix} \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}} \\ \eta_{\xi_{v'+1}}, \eta_{\xi_{v'+2}} \dots \eta_{\xi_n} \end{bmatrix}},$$

where P represents any function symmetrical in respect of $h_{q_1}, h_{q_2} \dots h_{q_v}$, and also in respect of $\eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_{v'}}$ (the interchanges, that is to say, between one h and another h , or between one η and another η , leaving P unaltered), it might be shown that the value of $\mathfrak{S}_{v,v'}$ resulting from the introduction of this more general value of R would (as for the particular value assumed) always be expressible as an integral function of the roots; and consequently, if P be taken of the same dimensions in the roots as the numerator of R previously assumed, that is vv' , $\mathfrak{S}_{v,v'}$ would continue to be (unless indeed it vanish) identical (to some numerical factor *près*) with the corresponding simplified residue. If, on the other hand, P be taken of less than vv' dimensions in h and η , we know *à priori* that $\mathfrak{S}_{v,v'}$ must vanish, as otherwise we should have a conjunctive of a weight less than the minimum weight. When P is of the proper amount of weight vv' , it is I think probable that another condition as to the distribution of the weight will be found to be necessary in order that $\mathfrak{S}_{v,v'}$ may not vanish, namely, that the highest power of any single h in P shall not exceed v , nor the highest power of any single η exceed v' . But as I have not had leisure to enter upon the inquiry, the verification or disproval of this supposed law, and more generally the evolution of the allotrious numerical factor introduced into $\mathfrak{S}_{v,v'}$ by assigning any particular form to P satisfying the necessary conditions of amount and distribution of weight, must be reserved, amongst other points connected with the theory of the remarkable forms (19) Art. 15, as a subject for future investigation.

Art. (31). A property of continued fractions, which, if known, I have not met with in any treatise on the subject (but which has been already cursorily alluded to in these pages), gives rise to a remarkable property of reciprocity connecting τ and t severally with $\mathfrak{S}$ in the syzygetic equation $\tau f - t\phi + \mathfrak{S} = 0$.

Let the successive convergents to the ordinary continued fraction

$$\frac{1}{q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \cdots + \frac{1}{q_{i-1} + \frac{1}{q_i}}}}}$$

be called

$$\frac{l_1}{m_1}, \quad \frac{l_2}{m_2} \cdots \frac{l_{i-1}}{m_{i-1}}, \quad \frac{l_i}{m_i},$$

respectively; it is well known that

$$m_{i-1}l_i - m_i l_{i-1} = (-)^{i-1} 1;$$

p. 480

but I believe that it has not been observed that this is only the extreme case of a much more general equation, namely

$$m_{i-\rho}l_i - m_i l_{i-\rho} = (-)^{i-\rho} \mu_{\rho-1},$$

where $\mu_1, \mu_2 \dots \mu_i$ denote respectively the denominators to the convergents to the continued fractions formed with the quotients taken in a reverse order, that is, the continued fraction

$$\frac{1}{q_i + \frac{1}{q_{i-1} + \frac{1}{q_{i-2} + \cdots + \frac{1}{q_2 + \frac{1}{q_1}}}}}$$

This is easily proved when $\rho = 1$; μ_0 is of course (as usual) to be considered 1. So more simply for the improper continued fraction,

$$\frac{l_i}{m_i} = \frac{1}{q_1 - \frac{1}{q_2 - \cdots - \frac{1}{q_{i-1} - \frac{1}{q_i}}}}$$

of which the convergents are supposed to be

$$\frac{l_1}{m_1}, \quad \frac{l_2}{m_2} \cdots \frac{l_{i-1}}{m_{i-1}}, \quad \frac{l_i}{m_i},$$

and the reverse fraction

$$\frac{1}{q_i - \frac{1}{q_{i-1} - \cdots - \frac{1}{q_2 - \frac{1}{q_1}}}},$$

of which the convergents are supposed to be

$$\frac{\lambda_1}{\mu_1}, \quad \frac{\lambda_2}{\mu_2}, \quad \frac{\lambda_i}{\mu_i},$$

we have the more simple equation

$$l_i m_{i-\rho} - l_{i-\rho} m_i + \mu_{\rho-1} = 0.$$

And it is well known, or at all events easily demonstrable, that

$$\frac{l_{i-1}}{l_i} = \frac{1}{q_i - \frac{1}{q_{i-1} - \frac{1}{q_{i-2} - \cdots - \frac{1}{q_2}}}}, \quad \frac{m_{i-1}}{m_i} = \frac{1}{q_i - \frac{1}{q_{i-1} - \frac{1}{q_{i-2} - \cdots - \frac{1}{q_2 - \frac{1}{q_1}}}}}.$$

Art. (32). If now we use subscript indices to denote the degree in x of the quantities to which they are affixed, we have the general syzygetic equation

$$K\tau_{n-\iota-1}f_m - Kt_{m-\iota-1}\phi_n + K\mathfrak{S}_\iota = 0,$$

where K , a constant (which I have given the means of determining in the first section), being rightly assumed, $K\tau_{n-\iota-1}, Kt_{m-\iota-1}$ become the numerator and denominator respectively of one of the convergents to $\frac{\phi}{f}$, expressed as p. 481

an improper continued fraction, and $K\mathfrak{S}_\iota$ becomes the denominator to one of the convergents to $\frac{\tau_{n-1}}{f}$, or, which is the same thing, to $\frac{\tau_{n-1}}{\phi}$. Conversely, it is obvious that if we adopt as our primitive functions cf_m and t_{m-1} , c being the value of K when $\iota = 0$, we shall obtain as the general form of our syzygetic equation, bearing in mind that $(m-1)$ now replaces n ,

$$cK\tau_{m-\iota-1}f_m - K'\mathfrak{S}_{m-\iota-1}t_{m-1} + K't_\iota = 0;$$

and similarly, if we adopt as our primitive functions τ_{n-1} and $c\phi_n$, we obtain for our general syzygetic equation, observing that $(n-1)$ now replaces m ,

$$K'\mathfrak{S}_{n-l-1}\tau_{n-1} - cK't_{n-l-1}\phi_n + K'\tau_l = 0;$$

so that (making abstraction of the constant factors and looking merely to the forms of the several functions which enter into the equations) we see that on the first hypothesis, namely of t_{m-1} being substituted for ϕ_n , the conjunctives of each degree in x change places with the second conjunctive factors, that is the original multipliers of ϕ of the same degree in x , and *vice versa*; and in the second hypothesis, where τ_{n-1} takes the place of f_m , the conjunctives of each degree in x change places with the first conjunctive factors, that is the original multipliers of f of the same degree in x , and *vice versa*; t_{m-1} and τ_{n-1} being respectively multipliers of ϕ and f , such that the difference of the respective products is independent of x . These results ought to be capable of being verified by aid of our general formulæ for $t, \tau, \mathfrak{S}$, and as this verification will serve to exhibit in a clearer light the nature of the reciprocity between the conjunctives and the conjunctive factors, it may be not uninteresting to set it out.

Art. (33). As usual, let $h_1, h_2 \dots h_m$ be the roots of fx , and $\eta_1, \eta_2 \dots \eta_n$ the roots of ϕx ; the last conjunctive factor to ϕ , which is of the degree $(m-1)$ in x , will be represented, neglecting powers of $(-)$, by t_{m-1} , where

$$t_{m-1} = \Sigma(x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_{m-1}}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_{m-1}} \\ \eta_1, \eta_2 \dots \eta_n \end{bmatrix}}{\begin{bmatrix} h_{q_m} \\ h_{q_1}, h_{q_2} \dots h_{q_{m-1}} \end{bmatrix}}.$$

If now we for greater simplicity make $t_{m-1} = t(x)$, and call the roots of t , $\eta'_1, \eta'_2 \dots \eta'_{m-1}$, any such quantity as

$$\begin{aligned} \left[\begin{array}{c} h_{q_m} \\ \eta'_1, \eta'_2 \dots \eta'_{m-1} \end{array} \right] &= t(h_{q_m}) = (h_{q_m} - h_{q_1})(h_{q_m} - h_{q_2}) \dots (h_{q_m} - h_{q_{m-1}}) \frac{\phi(h_{q_1})\phi(h_{q_2}) \dots \phi(h_{q_{m-1}})}{(h_{q_m} - h_{q_1})(h_{q_m} - h_{q_2}) \dots (h_{q_m} - h_{q_{m-1}})} \\ &= \phi(h_{q_1})\phi(h_{q_2}) \dots \phi(h_{q_{m-1}}) = R \frac{1}{\phi(h_{q_m})}, \end{aligned}$$

R denoting a constant independent of the root h_{q_m} selected, in fact the resultant of the two functions fx and ϕx , that is to say,

$$\phi(h_1)\phi(h_2)\phi(h_3) \dots \phi(h_m).$$

But by our general formula the simplified residue to fx and $t(x)$ of the ι th degree in x will be represented by

$$\mathfrak{S}'_{\iota,0} = \Sigma(x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_\iota}) \left\{ \frac{\begin{bmatrix} h_{q_{\iota+1}}, h_{q_{\iota+2}} \dots h_{q_m} \\ \eta'_1, \eta'_2 \dots \eta'_{m-1} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\iota} \\ h_{q_{\iota+1}}, h_{q_{\iota+2}} \dots h_{q_m} \end{bmatrix}} \right\};$$

therefore

$$\begin{aligned}\mathfrak{S}'_{\iota,0} &= \Sigma(x-h_{q_1})(x-h_{q_2})\cdots(x-h_{q_\iota}) \times \left\{ R^{m-\iota} \frac{\phi(h_{q_{\iota+1}})^{-1}\phi(h_{q_{\iota+2}})^{-1}\cdots\phi(h_{q_m})^{-1}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\iota} \\ h_{q_{\iota+1}}, h_{q_{\iota+2}} \dots h_{q_m} \end{bmatrix}} \right\} \\ &= R^{m-\iota-1} \Sigma(x-h_{q_1})(x-h_{q_2})\cdots(x-h_{q_\iota}) \frac{\phi(h_{q_1})\phi(h_{q_2})\cdots\phi(h_{q_\iota})}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\iota} \\ h_{q_{\iota+1}}, h_{q_{\iota+2}} \dots h_{q_m} \end{bmatrix}},\end{aligned}$$

or

$$\mathfrak{S}'_\iota = R^{m-\iota-1}t_\iota,$$

the relation which was to be obtained. So conversely, in precisely the same manner, calling t'_ι the conjunctive factor of the degree ι in x to $t(x)$ in the syzygetic equation which connects fx and $t(x)$ with a corresponding simplified residue, we have

$$\begin{aligned}t'_\iota &= \Sigma(x-h_{q_1})(x-h_{q_2})\cdots(x-h_{q_\iota}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\iota} \\ \eta'_1, \eta'_2 \dots \eta'_{m-1} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\iota} \\ h_{q_{\iota+1}}, h_{q_{\iota+2}} \dots h_{q_m} \end{bmatrix}} \\ &= R^{\iota-1} \Sigma(x-h_{q_1})(x-h_{q_2})\cdots(x-h_{q_\iota}) \frac{\phi(h_{q_{\iota+1}})\phi(h_{q_{\iota+2}})\cdots\phi(h_{q_m})}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_\iota} \\ h_{q_{\iota+1}}, h_{q_{\iota+2}} \dots h_{q_m} \end{bmatrix}} = R^{\iota-1}\mathfrak{S}_\iota,\end{aligned}$$

the conjugate equation to the one previously obtained¹.

And evidently the same reasoning serves to establish the reciprocity, or rather reciprocal convertibility, between the $\mathfrak{S}$ series and the τ series, when in lieu of the original primitives fx and ϕx we take as our primitives $\tau(x)$ and ϕx , $\tau(x)$ being the function which satisfies the equation

$$\tau(x)fx - t(x)\phi x + \mathfrak{S} = 0.$$

p. 483

Art. (34). It may be remarked that if $n = m - 1$, the last syzygetic equation being thus

$$t_{m-2}\phi_{m-1} - \tau_{m-2}f_m + \mathfrak{S}_0 = 0,$$

when t_{m-2} and f_m are taken as the primitives, the corresponding equation will be of the form

$$\mathfrak{S}'_{m-1}t_{m-2} - \tau'_{m-2}f_m + \mathfrak{S}'_0 = 0;$$

¹M. Hermite, by a peculiar method, first discovered one of these two conjugate relations of reciprocity, applicable to the case of Sturm's theorem, where $\phi x = f'x$, and I am indebted to him for bringing the subject under my notice.

these two equations must therefore be identical, and consequently $t_{m-2} = \phi_{m-1}$ (to a numerical factor *près*), so that t_{m-2} and ϕ_{m-1} are reciprocal forms. This is also obvious from the consideration that t_{m-2} must, by the general law of reciprocity (established above), be a residue to (f_m, ϕ_{m-1}) , which the latter function itself may be considered to be. Or the same fact may be presented under a more general point of view, by observing that when two functions are respectively the last conjunctive factors one to the other, the last conjunctive factor to either of them is the other itself; and it is an immediate corollary that, if ϕ is the differential derivative of f , the last conjunctive factor to f is the differential derivative of f itself.

SECTION III.

On the application of the Theorems in the preceding Section to the expression in terms of the roots of any auxiliary functions of Sturm's auxiliary functions, and the other functions which connect them with the primitive function and its first differential derivative.

Art. (35). The formulæ in the preceding Section had reference to the case of two absolutely independent functions and their respective systems of roots; when the functions become so related that the roots of the one system become explicitly or implicitly functions of the roots of the other system, the formulæ will become susceptible in some of these latter also, and, in some cases the terms of which the sums are taken will vanish by reason of numerator factors becoming zero, in other cases terms will require to be evaluated by means of ultimate ratios; but, as I shall proceed to show, in the case here treated of, which now p. 484

is the differential derivative of the other. When f and ϕ are thus related, so that $\phi = \frac{df}{dx}$, calling as before $h_1, h_2 \dots h_m$ the roots of f , and $\eta_1, \eta_2 \dots \eta_{m-1}$ the roots of ϕ , we shall have in general

$$\begin{aligned} & \left[\begin{array}{c} h_{q_{i+1}} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{array} \right] = (h_{q_{i+1}} - \eta_1)(h_{q_{i+1}} - \eta_2) \dots (h_{q_{i+1}} - \eta_{m-1}) \\ & = f'_{h_{q_{i+1}}} = \left[\begin{array}{c} h_{q_{i+1}} \\ h_{q_1}, h_{q_2} \dots h_{q_i}, h_{q_{i+2}} \dots h_{q_m} \end{array} \right] = \left[\begin{array}{c} h_{q_{i+1}} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{array} \right] \times \left[\begin{array}{c} h_{q_{i+1}} \\ h_{q_{i+2}}, h_{q_{i+3}} \dots h_{q_m} \end{array} \right]. \end{aligned}$$

Consequently

$$\begin{aligned}
\begin{bmatrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{bmatrix} &= \begin{bmatrix} h_{q_{i+1}} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{bmatrix} \text{imes} \begin{bmatrix} h_{q_{i+2}} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{bmatrix} \times \dots \\
&\times \begin{bmatrix} h_{q_m} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{bmatrix} \\
&= \begin{bmatrix} h_{q_{i+1}} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{bmatrix} \text{imes} \begin{bmatrix} h_{q_{i+1}} \\ h_{q_{i+2}}, h_{q_{i+3}} \dots h_{q_m} \end{bmatrix} \\
&\times \begin{bmatrix} h_{q_{i+2}} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{bmatrix} \text{imes} \begin{bmatrix} h_{q_{i+2}} \\ h_{q_{i+1}}, h_{q_{i+3}} \dots h_{q_m} \end{bmatrix} \\
&\times \dots \times \begin{bmatrix} h_{q_m} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{bmatrix} \text{imes} \begin{bmatrix} h_{q_m} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_{m-1}} \end{bmatrix}.
\end{aligned}$$

Hence

$$\frac{\begin{bmatrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{bmatrix}}{\begin{bmatrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{bmatrix}} = \prod_{r=i+1}^m \begin{bmatrix} h_{q_r} \\ h_{q_{i+1}}, \dots, h_{q_r}, \dots, h_{q_m} \end{bmatrix} = (-)^{\frac{1}{2}(m-i)(m-i-1)} \zeta(h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m})$$

the ζ denoting the operation of taking the product of the squares of the differences of the quantities which this symbol governs. Hence the Bezoutian secondary to f and f' of the $(m-i-1)$ th degree in x , namely

$$(-)^{\frac{i(i-1)}{2}} \Sigma(x-h_{q_{i+1}})(x-h_{q_{i+2}}) \dots (x-h_{q_m}) \frac{\begin{bmatrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}},$$

becomes

$$(-)^{i(m-i)} \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_i})(x-h_{q_{i+1}})(x-h_{q_{i+2}}) \dots (x-h_{q_m}) = \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_i})(x-h_{q_{i+1}}) \dots (x-h_{q_m})$$

p. 485

since $(-)^{i(m-i)} = 1$; this gives the well-known formulæ (enunciated² by me in the *London and Edinburgh Philosophical Magazine* for 1839) for expressing M. Sturm's auxiliary functions in terms of the roots of the primitive, and which I therein stated were immediately deducible from the general formulæ (also enunciated in the same paper) applicable to any two functions. These more general formulæ appear to have completely escaped the notice of M. Sturm and others, who have used the special formulæ applicable to the case of one function becoming the first differential derivative of the other.

²p. 45 above.

Art. (36). In precisely the same manner, if we form as usual the ordinary syzygetic equation

$$tf'x - \tau fx + \mathfrak{S} = 0,$$

we may find the different values of t given by the complementary formulæ; and using t_i to denote the multiplier of the degree i in x , that is appertaining to the residue of the degree $(m - i - 1)$ in x , we have

$$\begin{aligned} t_i &= \Sigma \frac{\left[\begin{array}{c} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{array} \right]}{\left[\begin{array}{c} h_{q_1}, h_{q_2} \dots h_{q_i} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \end{array} \right]} (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_i}) \\ &= \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_i})(x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_i}). \end{aligned}$$

Art. (37). Thus, if we make $i = m - 1$,

$$f'x = t_{m-1} = \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}})(x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_{m-1}}).$$

It is evident from the form of $f'x$ that it possesses relative to fx , the same property as $f'x$, I mean the property that when x is indefinitely near to a real root of fx , and is passing from the inferior to the superior side of such root, $\frac{f'x}{fx}$ like $\frac{f'x}{fx}$ will pass from being negative to being positive, or in other words, f'_1x and $f'x$ have always the same sign in the immediate vicinity to a real root of fx . Hence it follows that f'_1x might be used instead of $f'x$, to produce, by the Sturmian process of common measure, a series of auxiliary functions, which with fx and f'_1x would form a rhizoristic series, that is a series for determining (as in the manner of M. Sturm's ordinary auxiliaries) the number of real roots of fx comprised within given limits. The rhizoristic series generated by this process will, it is easily seen, be (to a constant factor *près*) the denominators (reckoning +1 as the denominator in the zero place) of the successive convergents to $\frac{f'x}{fx}$ thrown under the form

p. 486

of a continued fraction

$$\cfrac{1}{q_1 - \cfrac{1}{q_2 - \dots - \cfrac{1}{q_{n-1} - \cfrac{1}{q_n}}}};$$

M. Sturm's own rhizoristic series, on the contrary, will be (to a constant factor *près*) the denominators of the convergents to the inverse fraction $\frac{f'x}{fx}$, which will

be of the form

$$K \left(\begin{array}{c} 1 \\ \hline 1 \\ q_n - \frac{1}{\hline q_{n-1} - \cdots - \frac{1}{\hline q_2 - \frac{1}{q_1}}} \end{array} \right);$$

accordingly these two rhizoristic series will be equivalent as regards the number of changes and of combinations of sign (afforded by each) corresponding to any given value of x , of which of course the q 's are linear functions. This result agrees with what has been demonstrated by me³ by a more general method (in the *London and Edinburgh Philosophical Magazine*, June and July 1853), where it has been proved, by means of a very simple theorem of determinants, that the two series

$$\frac{1}{q_1}, \quad \frac{1}{q_1} - \frac{1}{q_2}, \quad \frac{1}{q_1} - \frac{1}{q_2} - \frac{1}{q_3}, \quad \dots, \quad \frac{1}{q_1} - \frac{1}{q_2} - \frac{1}{q_3} - \dots - \frac{1}{q_n},$$

and

$$\frac{1}{q_n}, \quad \frac{1}{q_n} - \frac{1}{q_{n-1}}, \quad \frac{1}{q_n} - \frac{1}{q_{n-1}} - \frac{1}{q_{n-2}}, \quad \dots, \quad \frac{1}{q_n} - \frac{1}{q_{n-1}} - \frac{1}{q_{n-2}} - \dots - \frac{1}{q_1},$$

always contain (for real values of $q_1, q_2, q_3 \dots q_n$) the same number of positive and negative signs.

Art. (38). Having now determined the general values of $\mathfrak{S}$ and t in the equation $tf'x - \tau fx + \mathfrak{S} = 0$ as explicit integral functions of the roots of fx , the more difficult task remains to assign to τ its value similarly expressed. This cannot readily be effected by means of substitutions in the general formulæ, the method we adopted for finding t and $\mathfrak{S}$; but all the other quantities except τ in the syzygetic equation being integral functions of the roots, it is evident that τ also must be an integral function of the same, and to obtain it we may use the expression

$$\tau = \frac{tf'x + \mathfrak{S}}{fx}.$$

To obtain the general form of τ by direct calculation from this formula would however be found to be impracticable; the mode I adopt therefore to discover the general expression for τ corresponding to different values of $\mathfrak{S}$, is to ascertain its value on the hypothesis of particular relations existing between the roots of fx , and then from the particular values of τ thus obtained to infer demonstratively

³See below pp. 616 and 621.

its general form, as will be seen below. The demonstration of τ is unavoidably somewhat long, τ being in fact represented by a double sum of partial symmetrical functions.

Using the subscript indices of each function as the syzygetic equation to denote its degree in x , we have in general

$$t_{m-i-1}f'x - \tau_{m-i-2}fx + \mathfrak{S}_i = 0,$$

p. 487

where if we make

$$h_1 - x = k_1, quad h_2 - x = k_2, \dots, h_m - x = k_m,$$

so that

$$h_i - h_\omega = k_i - k_\omega,$$

and therefore

$$\zeta(h_{\theta_1}, h_{\theta_2} \dots h_{\theta_\rho}) = \zeta(k_{\theta_1}, k_{\theta_2} \dots k_{\theta_\rho}),$$

we have in effect found

$$\mathfrak{S}_i = \Sigma k_{q_1} k_{q_2} \dots k_{q_i} \zeta(k_{q_{i+1}}, k_{q_{i+2}} \dots k_{q_m})$$

and

$$t_{m-i-1} = \pm \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-i-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-i-1}});$$

we have also $f'(x) = (-)^{m-1} \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}}$.

Let us commence with the case where $i = 0$; we have then

$$\mathfrak{S}_0 = \zeta(k_1, k_2 \dots k_m),$$

$$t_{m-1} = \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}});$$

we have thus

$$(-)^m \tau_{m-2} k_1 k_2 \dots k_m = \zeta(k_1, k_2 \dots k_m) - \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \times \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}})$$

It may easily be verified that the negative sign interposed between the two parts of the right-hand member of the equation has been correctly taken, for

$$\zeta(k_1, k_2 \dots k_m) \text{ contains a term } k_1^{2(m-1)} k_2^{2(m-2)} \dots k_{m-2}^4 k_{m-1}^2,$$

$$\Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \text{ contains a term } k_1 k_2 \dots k_{m-2} k_{m-1},$$

and

$$\Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}})$$

contains a term $k_1^{2m-3} k_2^{2m-5} \dots k_{m-2}^3 k_{m-1}$, and thus the term $k_1^{2(m-1)} k_2^{2(m-2)} \dots k_{m-2}^4 k_{m-1}^2$, which does not contain k_m , will (as it ought to do) disappear from the right-hand side of the equation.

Now suppose

$$k_1 = k_2,$$

then

$$\zeta(k_1, k_2 \dots k_m) = 0,$$

and also

$$\zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}) = 0,$$

except when one or the other of the two disjunctive equations

$$q_1, q_2, q_3 \dots q_{m-1} = 1, 3, 4 \dots m,$$

$$q_1, q_2, q_3 \dots q_{m-1} = 2, 3, 4 \dots m,$$

is satisfied (by a disjunctive equation, meaning an equation which affirms the equality of one set of quantities with another set the same in number, each with each, but in some unassigned order).

p. 488

Hence

$$\Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}) = 2k_1 k_3 \dots k_m \zeta(k_1, k_3 \dots k_m).$$

Hence when $k_1 = k_2$, $(-)^m \tau_{m-2}$ becomes

$$\frac{2}{k_1} \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_1, k_3 \dots k_m),$$

that is

$$2\zeta(k_1, k_3 \dots k_m) \{k_1 \Sigma k_{r_3} k_{r_4} \dots k_{r_{m-1}} + 2k_3 k_4 \dots k_m\},$$

the Σ referring to $r_3, r_4 \dots r_m$ supposed to be disjunctively equal to $3, 4 \dots m$.

Now τ_{m-2} is of $(m-2)$ dimensions in x , and whenever more than one equality exists between the h 's, $\mathfrak{S}_0$ and t_{m-1} both vanish (in fact every term in each vanishes separately), and therefore τ_{m-2} , which

$$= \frac{\mathfrak{S}_0 + t_{m-1} f'x}{k_1 k_2 \dots k_m},$$

will vanish. Hence $(-)^m \tau_{m-2}$ must be always of the form

$$\Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}}) \times \Psi(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}, k_{q_m}),$$

Ψ denoting some integral function of $(m-2)$ dimensions in respect of the system of quantities $k_{q_1}, k_{q_2} \dots k_{q_m}$. The result above obtained enables us to assign the value of

$$\Psi(k_1, k_3 \dots k_m, k_2),$$

when $k_1 = k_2$, namely

$$k_1 \Sigma(k_{r_3}, k_{r_4} \dots k_{r_{m-1}}) + 2k_3 k_4 \dots k_m.$$

Now for a moment suppose, selecting $(m - 1)$ terms $k_1, k_3, k_4 \dots k_m$ out of the m terms of the k series, that

$$\Omega(k_1, k_3, k_4 \dots k_m, k_2) = k_2^{m-2} - k_2^{m-3} S_1(k_1, k_3 \dots k_m) + k_2^{m-4} S_2(k_1, k_3 \dots k_m) \\ \pm \dots + k_2 S_{m-3}(k_1, k_3 \dots k_m) \pm 2 S_{m-2}(k_1, k_3 \dots k_m),$$

where S_1 means that the quantities which it governs are to be simply added together, S_2 denotes that their binary, S_3 that their ternary, and in general S_r that their r -ary products are to be added together.

When $k_1 = k_2$, Ω becomes

$$k_1^{m-2} - k_1^{m-3} \{k_1 + S_1(k_3, k_4 \dots k_m)\} + k_1^{m-4} \{k_1 S_1(k_3, k_4 \dots k_m) + S_2(k_3, k_4 \dots k_m)\} \\ - k_1^{m-5} \{k_1 S_2(k_3, k_4 \dots k_m) + S_3(k_3, k_4 \dots k_m)\} \pm \dots \\ \pm k_1 \{k_1 S_{m-4}(k_3, k_4 \dots k_m) + S_{m-3}(k_3, k_4 \dots k_m)\} \pm 2 S_{m-2}(k_3, k_4 \dots k_m),$$

which evidently equals

$$\pm \{2 S_{m-2}(k_3, k_4 \dots k_m) + k_1 S_{m-3}(k_3, k_4 \dots k_m)\},$$

that is

$$\pm \{k_1 \Sigma(k_{r_3}, k_{r_4} \dots k_{r_{m-1}}) + 2 k_3 k_4 \dots k_m\}.$$

p. 489

Hence when $k_1 = k_2$, $\Psi = \Omega$, and

$$(-)^m \tau_{m-2} = \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}}) \text{imes} \Omega(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}, k_{q_m});$$

and so in like manner, when h_1 is equal to any one of the $(m - 1)$ quantities $k_2, k_3 \dots k_m$, the form of τ_{m-2} above written will have been correctly assumed. But τ_{m-2} may be treated as a function of $(m - 2)$ dimensions in k_1 , and consequently any form of $(m - 2)$ dimensions in k_1 , which fits it for $(m - 1)$ different values of k_1 , must be its general form, and accordingly we have universally,

$$(-)^m \tau_{m-2} = \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}}) \times \{(x - h_{q_m})^{m-2} \pm (x - h_{q_m})^{m-3} S_1(x - h_{q_1}, x - h_{q_2} \dots \\ + (x - h_{q_m})^{m-4} S_2(x - h_{q_1}, x - h_{q_2} \dots x - h_{q_{m-1}}) + \&c. \\ \mp (x - h_{q_m}) S_{m-3}(x - h_{q_1}, x - h_{q_2} \dots x - h_{q_{m-1}}) \\ \pm 2 S_{m-2}(x - h_{q_1}, x - h_{q_2} \dots x - h_{q_{m-1}})\}.$$

Art. (39). With a view to better paving our way to the general form of τ for all values of i , let us pass over the case of $i = 1$ and go at once to the equation

$$t_{m-3} f' x - \tau_{m-4} f x + \mathfrak{S}_2 = 0;$$

and to better fix our ideas let $m = 7$, so that the equation becomes

$$t_4 f' x - \tau_3 f x + \mathfrak{S}_2 = 0;$$

we have then, preserving the same relation as before, that is, using h to denote any root of fx , and k to denote $h - x$, the equation

$$\pm k_1 k_2 k_3 k_4 k_5 k_6 k_7 \tau_3 = \Sigma k_{q_1} k_{q_2} \zeta(k_{q_3} k_{q_4} k_{q_5} k_{q_6} k_{q_7}) - \Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} k_{q_5} k_{q_6} \times \Sigma \{k_{q_1} k_{q_2} k_{q_3} k_{q_4} \zeta(k_{q_5} k_{q_6} k_{q_7})\}$$

now τ_3 will vanish whenever more than three relations of equality exist between the k 's, for then each term in *both* of the two sums in the right-hand member of the equation above written will separately vanish; and of course three relations of equality between the same are sufficient to make all the terms in the first of these sums vanish. This relationship between the different k 's corresponding to a multiplicity 3 may arise in different ways; the multiplicity 3 may be divided into 3 units corresponding to 3 pairs of equal roots, or into 2 and 1 corresponding one set of 3 equal roots, and a second set of 2 equal roots, or may be taken *en bloc*, which corresponds to the case of one set of 4 equal roots. I shall make the first of these suppositions, which will sufficiently well answer our purpose in the case before us.

Thus I shall suppose

$$k_1 = k_4, \quad k_2 = k_5, \quad k_3 = k_6,$$

then, as above remarked,

$$\zeta(k_{q_3} k_{q_4} k_{q_5} k_{q_6} k_{q_7}) = 0$$

p. 490

for all values of q_3, q_4, q_5, q_6, q_7 , and therefore

$$\Sigma k_{q_1} k_{q_2} \zeta(k_{q_3} k_{q_4} k_{q_5} k_{q_6} k_{q_7}) = 0;$$

also $\Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} k_{q_5} k_{q_6}$ becomes

$$k_1 k_2 k_3 \{k_1 k_2 k_3 + 2k_7(k_1 k_2 + k_1 k_3 + k_2 k_3)\},$$

and $\zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_4})$ vanishes, except for the cases where q_1, q_2, q_3, q_4 represent respectively, q_1 the index 1 or 4, q_2 the index 2 or 5, q_3 the index 3 or 6, and q_4 the index 7.

Hence

$$\Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} \zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_4}) = 2^3 k_1 k_2 k_3 k_7 \zeta(k_1 k_2 k_3 k_7),$$

and consequently τ_3 becomes

$$\pm 8 \zeta(k_1 k_2 k_3 k_7) \times \{k_1 k_2 k_3 + 2k_7(k_1 k_2 + k_1 k_3 + k_2 k_3)\}.$$

Hence we are able to predict that the general expression for our τ in the case before us will be

$$\begin{aligned} \tau_3 = & \mp \Sigma \zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_7}) \\ & \times \{(k_{q_4}^3 + k_{q_5}^3 + k_{q_6}^3) - (k_{q_4}^2 + k_{q_5}^2 + k_{q_6}^2)(k_{q_1} + k_{q_2} + k_{q_3} + k_{q_7}) \\ & + (k_{q_4} + k_{q_5} + k_{q_6})(k_{q_1} k_{q_2} + k_{q_1} k_{q_3} + k_{q_1} k_{q_7} + k_{q_2} k_{q_3} + k_{q_2} k_{q_7} + k_{q_3} k_{q_7}) \\ & - 4(k_{q_1} k_{q_2} k_{q_3} + k_{q_1} k_{q_2} k_{q_7} + k_{q_1} k_{q_3} k_{q_7} + k_{q_2} k_{q_3} k_{q_7})\}. \end{aligned}$$

For in the first place, the fact that the τ vanishes when more than three relations of equality exist between the k 's, proves that we may assume τ_3 of the form

$$\Sigma \zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_7}) \times \phi\{k_{q_1} k_{q_2} k_{q_3} k_{q_7}; k_{q_4} k_{q_5} k_{q_6}\},$$

the semicolon (;) separating the k 's into two groups, in respect of each of which severally ϕ is a symmetrical form. But if in the expression last above written for τ_3 we make

$$k_1 = k_4, quad k_2 = k_5, \quad k_3 = k_6,$$

it becomes

$$\mp 8\zeta(k_1 k_2 k_3 k_7) \times \{(k_1^3 + k_2^3 + k_3^3) - (k_1^2 + k_2^2 + k_3^2)(k_1 + k_2 + k_3 + k_7) \\ + (k_1 + k_2 + k_3)(k_1 k_2 + k_1 k_3 + k_1 k_7 + k_2 k_3 + k_2 k_7 + k_3 k_7) - 4(k_1 k_2 k_3 + k_1 k_2 k_7 + k_1 k_3 k_7 + k_2 k_3 k_7)\}.$$

Now in general if

$$\sigma_r = a_1^r + a_2^r + a_3^r + \dots + a_t^r,$$

and

$$S_r = \Sigma(a_1 a_2 a_3 \dots a_r),$$

then

$$\sigma_r - \sigma_{r-1} S_1 + \sigma_{r-2} S_2 \pm \dots \pm r S_r = 0.$$

Consequently the sum of the terms constituting the second factor in the above expression

$$= (3 - 4)k_1 k_2 k_3 + (2 - 4)k_7(k_1 k_2 + k_1 k_3 + k_2 k_3).$$

p. 491

Hence the above expression becomes

$$\pm 8\zeta(k_1 k_2 k_3 k_7) \{k_1 k_2 k_3 + 2(k_1 k_2 + k_1 k_3 + k_2 k_3)k_7\}.$$

Thus, then, whenever k_1, k_2, k_3 are respectively equal to any three of the quantities k_4, k_5, k_6, k_7 , which may take place in twenty-four different ways (twenty-four being the number of permutations of four things), our τ_3 will have been correctly assumed; but $\zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_7})$ being replaceable by $\zeta(h_{q_1} h_{q_2} h_{q_3} h_{q_7})$, the τ_3 may be treated as a cubic function in k_1, k_2, k_3 , and arranged according to the powers of k_1, k_2, k_3 will contain only twenty terms; hence, since the assumed form is verified for more than twenty, that is, for twenty-four values of h_1, h_2, h_3 , it follows that the assumed form is universally identical with the form of τ , which was to be determined.

Art. (40). Now, again, in order to facilitate the conception of the general proof, let us suppose $f x$ to be of only five dimensions in x , i still remaining 3; it will no longer be possible when we suppose a multiplicity three to prevail among the roots, to conceive this multiplicity to be distributed into three parts, for that would require the existence of three pairs of roots, there being only five. But we

may, if we please, make $h_1 = h_2 = h_3$, and $h_4 = h_5$, or else $h_1 = h_2 = h_3 = h_4$, or in any other mode conceive the multiplicity to be divided into two parts, 2 and 1 respectively, or to be taken collectively *en bloc*. As a mode of proceeding the more remote from that last employed, I shall choose the latter supposition. Then we obtain (τ now becoming τ_{5-2-2} , that is τ_1)

$$k_1 k_2 k_3 k_4 k_5 \tau_1 = \pm \Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} \times \Sigma k_{q_1} k_{q_2} \zeta(k_{q_1} k_{q_2}),$$

and $\zeta(k_{q_1} k_{q_2})$ will vanish, except in the case where q_1 represents the indices 1 or 2 or 3 or 4, and q_2 the index 5; also

$$\Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} = k_1^4 + 4k_1^3 k_5.$$

Hence our equation becomes

$$k_1^4 k_5 \tau = (k_1^4 + 4k_1^3 k_5) 4k_1 k_5 \zeta(k_1 k_5),$$

and τ becomes

$$-4\zeta(k_1 k_5)(k_1 + 4k_5).$$

If, now, we assume for the general value of τ in the case before us

$$\tau = \Sigma \zeta(k_{q_1} k_{q_5}) \{ (k_{q_2} + k_{q_3} + k_{q_4}) - 4(k_{q_1} + k_{q_5}) \},$$

when $k_1 = k_2 = k_3 = k_4$, τ becomes

$$\pm 4\zeta(k_1 k_5) \{ 3k_1 - (4k_1 + k_5) \},$$

that is

$$\pm 4\zeta(k_1 k_5)(k_1 + 4k_5).$$

Hence then for the two systems of values of h_1, h_2, h_3 , namely

$$\begin{array}{ll} h_1 = h_4 & h_1 = h_5 \\ h_2 = h_4 & \text{or} \quad h_2 = h_5 \\ h_3 = h_4 & h_3 = h_5, \end{array}$$

the form of τ will have been correctly assumed. But since the derived form is a linear function of h_1, h_2, h_3 , this is not enough to identify the assumed with the general form, since for such verification four systems of values must be taken, four being the number of terms in a function of three variables of the first degree. If, however, we had adopted a separation of the multiplicity three into two parts, and had started with supposing $k_1 = k_2 = k_3$, $k_4 = k_5$, we should have found that τ would have become

$$= 6\zeta(k_1, k_5)(2k_1 + 3k_5).$$

Moreover, when these equalities subsist,

$$k_1 k_2 k_3 k_4 + k_1 k_2 k_3 k_5 + k_1 k_2 k_4 k_5 + k_1 k_3 k_4 k_5 + k_2 k_3 k_4 k_5$$

becomes $2k_1^3 k_5 + 3k_1^2 k_5^2$, and the common factor $k_1^2 k_4$ disappears in the course of the operations for finding τ , and eventually we have to show (in order to support the universality of the previously assumed form for τ) that

$$k_{q_2} + k_{q_3} + k_{q_4} - 4(k_{q_1} + k_{q_5})$$

becomes $-2k_{q_4} - 3k_5$ when

$$k_{q_2} = k_{q_3} = k_{q_1} = k_1, \quad k_{q_4} = k_{q_5} = k_5,$$

which is evidently true. Hence then τ will have been correctly assumed for the following cases,

$$k_1 = k_2 = k_5 = k_3, \quad k_1 = k_2 = k_5 = k_4;$$

and also for the cases

$$\left. \begin{array}{l} k_1 = k_2 = k_3 \text{ and } k_5 = k_4 \\ k_1 = k_5 = k_3 \text{ and } k_2 = k_4 \\ k_2 = k_5 = k_3 \text{ and } k_1 = k_4 \end{array} \right\} \\ \left. \begin{array}{l} k_1 = k_2 = k_4 \text{ and } k_5 = k_3 \\ k_1 = k_5 = k_4 \text{ and } k_2 = k_3 \\ k_2 = k_5 = k_4 \text{ and } k_1 = k_3 \end{array} \right\},$$

that is, for eight cases in all, whereas four only would have sufficed. Hence, *ex abundantia demonstrationis*, the form assumed for τ_1 is in the case before us the general form.

Art. (41). We may now easily write down the general form which τ assumes for all values of i and prove its correctness. If the roots be

$$h_1, h_2, h_3 \dots h_m,$$

and

$$t_{m-i-1} f' x - \tau_{m-i-2} f x + \mathfrak{S}_i = 0,$$

p. 493

we shall have

$$\pm \tau_{m-i-2} = \Sigma \{ \zeta (h_{q_1} h_{q_2} h_{q_3} \dots h_{q_{m-i-1}}) \times [\sigma_{m-i-2} - \sigma_{m-i-3} S_1 + \sigma_{m-i-4} S_2 + \&c. \\ + (-)^{m-i-3} \sigma_1 S_{m-i-3} + (-)^{m-i-2} (\sigma_0 + 1) S_{m-i-2}] \},$$

where σ_r denotes in general the sum of the r th powers of the $(i+1)$ quantities

$$(x - h_{q_{m-i}}), (x - h_{q_{m-i+1}}), \dots (x - h_{q_m}),$$

and S_r denotes in general the sum of the products of the complementary $(m-i-1)$ quantities

$$(x - h_{q_1}), (x - h_{q_2}) \dots (x - h_{q_{m-i-1}})$$

combined r and r together. It will of course also be understood that $\sigma_0 = i + 1$, so that $\sigma_0 + 1 = i + 2$.

Art. (42). To prove the correctness of this general determination of the form of τ_{m-i-2} , let us suppose in general that $i+1$ relations of equality spring up between the m quantities $h_1, h_2 \dots h_m$; we shall then easily obtain (N representing a certain numerical multiplier)

$$\pm Q = N \zeta(k_1, k_2 \dots k_{m-i-1}) \frac{\Sigma k_{\theta_1} k_{\theta_2} \dots k_{\theta_{m-i-1}}}{\mu_1^{\mu_1-1} k_1^{\mu_2-1} \dots \mu_{m-i-1}^{\mu_{m-i-1}-1}},$$

$k_1, k_2 \dots k_{m-i-1}$ being what the k system becomes when repetitions are excluded, and being respectively supposed to occur $\mu_1, \mu_2 \dots \mu_{m-i-1}$ times respectively, so that

$$\mu_1 + \mu_2 + \dots + \mu_{m-i-1} = m;$$

the fractional part of the right-hand member of the equation immediately above written will be readily seen to be equivalent to

$$\Sigma \mu_{\theta_{m-i-1}} k_{\theta_1} k_{\theta_2} \dots k_{\theta_{m-i-2}}.$$

To establish the correctness of the assumed form, we must be able, as in the particular cases previously selected, to prove two things; the one, and the more difficult thing to be proved is, that when the series of distinct quantities $k_1, k_2, k_3 \dots k_m$ become converted into μ_1 groups of k_1 ; μ_2 groups of k_2 , $\dots \mu_{m-i-1}$ groups of k_{m-i-1} , then that

$$\Sigma \mu_{\theta_1} k_{\theta_2} k_{\theta_3} k_{\theta_4} \dots k_{\theta_{m-i-1}},$$

or in other terms

$$\Sigma \pm k_{\theta_1} k_{\theta_2} k_{\theta_3} \dots k_{\theta_{m-i-1}} \times \sum_{\pi=1}^{m-i-1} (\mu_{\theta}),$$

becomes identical with

$$\sigma_{m-i-2} - \sigma_{m-i-3} S_1 + \&c. + (-1)^{m-i-2} (\sigma_0 + 1) S_{m-i-2}.$$

The other step to be made, and with which I shall commence, consists in showing that the number of terms in the expression last above written, considered as a function of $(m-i-2)$ th degree of $(i+1)$ variables, is never greater than the entire number of ways in which $(i+1)$ quantities out of m quantities may be equated to the remaining $(m-i-1)$ quantities, namely each of the first set respectively to all the same, or all different, or some the